

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 5/1/2025

Compounds: Amoxicillin, Flunitrazepam

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Amoxifluazepam

Formula: C₂₃H₂₆ClN₃O₄

Weight: 443.94

Drug-likeness: 0.65

Activity:

Antibiotic and Anxiolytic

Target Analysis

Penicillin-binding Protein

Binding Affinity: -8

Essential for bacterial cell wall synthesis

GABA-A Receptor

Binding Affinity: -9.5

Modulates inhibitory neurotransmission, enhancing GABA effect

Toxicity Profile

Risk Level: Moderate

Affected Systems: Liver, Central Nervous System, Kidneys

Mitigation Suggestions:

- Monitor liver function tests regularly
- Avoid in patients with existing CNS disorders

Pharmacokinetics

Absorption:

Moderate oral bioavailability

Distribution:

Widely distributed in body fluids and tissues

Metabolism:

Hepatic metabolism primarily via CYP3A4

Excretion:

Renal elimination with some biliary clearance

Half-life: 6-10 hours

Detailed Analysis

Amoxifluazepam is a hypothetical bifunctional pharmaceutical molecule synthesized by integrating key functional moieties of amoxicillin and flunitrazepam. The combination leverages the beta-lactam core of amoxicillin to disrupt bacterial cell wall biosynthesis while harnessing the benzodiazepine structure of flunitrazepam to potentiate GABA-A receptor activity, imparting sedative and anxiolytic benefits. This dual mechanism is particularly strategic for therapeutic interventions in complex infections with neuropsychiatric components. The compound's molecular backbone and side chains are optimized through computational chemistry for enhanced binding targets, yielding significant binding energies and predicting reasonable selectivity for penicillin-binding proteins and GABA-A receptors. Nonetheless, due consideration is required for its moderate toxicity profile, necessitating well-devised clinical protocols to navigate potential hepatotoxic and CNS risks. Its pharmacokinetic parameters reveal a balanced absorption and metabolism profile, mediated by cytochrome P450 enzymes, with a half-life conducive for manageable dosing regimens. Future prospects entail progressive clinical validation and application diversification, positing Amoxifluazepam as a promising contender in therapeutics for infectious diseases with psychiatric manifestations.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>